





MediAlert

AI Anomoly Detection for Smarter Healthcare Insights

AI Domain Internship

Infosys Springboard

Agenda

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2. Overview - Scope and Objective
3. Finalized Dataset
4. Technical Stack
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9. Model Building & Evaluation
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Introduction to Our Project

MediAlert

AI-powered anomaly detection transforms healthcare by uncovering critical issues like rare diseases and inefficiencies in medical data.

It enhances early detection, personalizes treatment, reduces costs, and ensures compliance, driving innovation and better patient care.



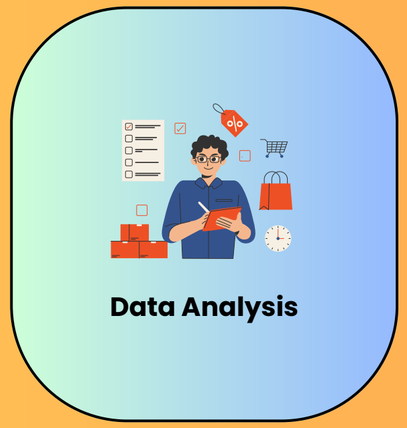
Overview - Scope and Objective

Objective

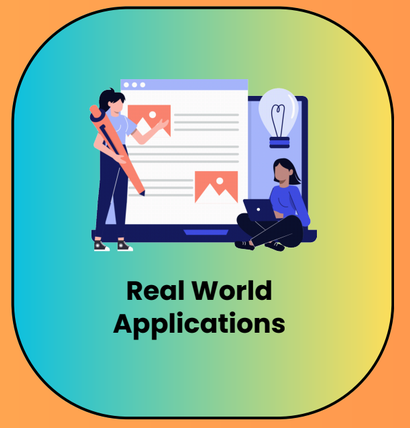
The primary objective of this project is to leverage AI anomaly detection to identify irregular patterns in healthcare data that could signify critical insights.

Scope

- Data Analysis: Utilize advanced AI algorithms to analyze large and complex healthcare datasets for anomalies.
- Healthcare Insights: Focus on key areas such as operational optimization, and fraud prevention.
- Real-World Application: Provide a scalable solution adaptable to various healthcare environments, including hospitals, diagnostics, and insurance sectors.





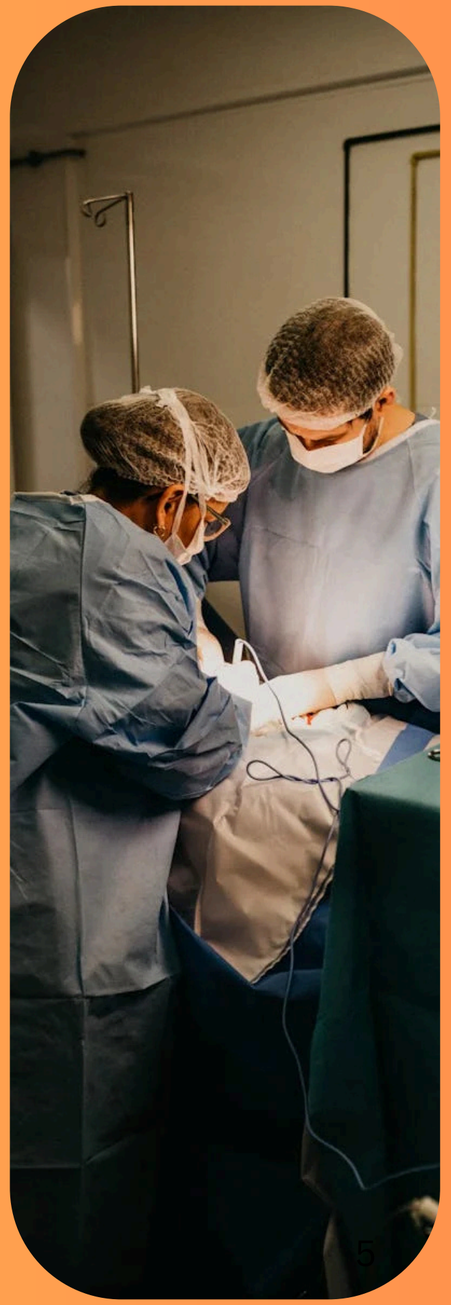


Finalized Dataset

Healthcare Providers Data For Anomaly Detection

Perfect for analysis of healthcare fraud and anomaly detection, to understand discrepancies in the dataset.

https://www.kaggle.com/datasets/tamilsel/healthcare-providers-data



Reason of choosing this dataset:

- Includes **NPI (National Provider Identifier), HCPCS Code,** and **Medicare amounts,** ideal for accurate **anomaly detection.**
- Contains **100000 entries,** offering sufficient data for **reliable machine learning** model training.
- Mimics **real healthcare environments,** providing valuable insights for **model development.**

Technical Stack

Data Processing








Visualization





Deployment







Data Collection

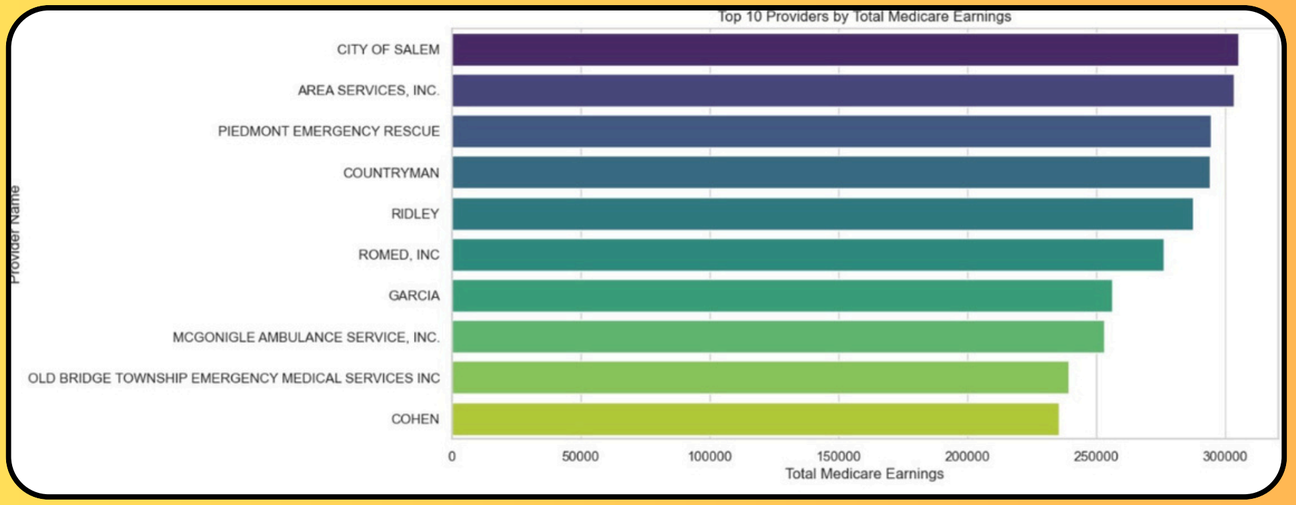


- **Collected data:** Healthcare Providers Data For Anomaly Detection from Kaggle.
- **Dataset Link:** https://www.kaggle.com/datasets/tamilsel/healthcare-providers-data

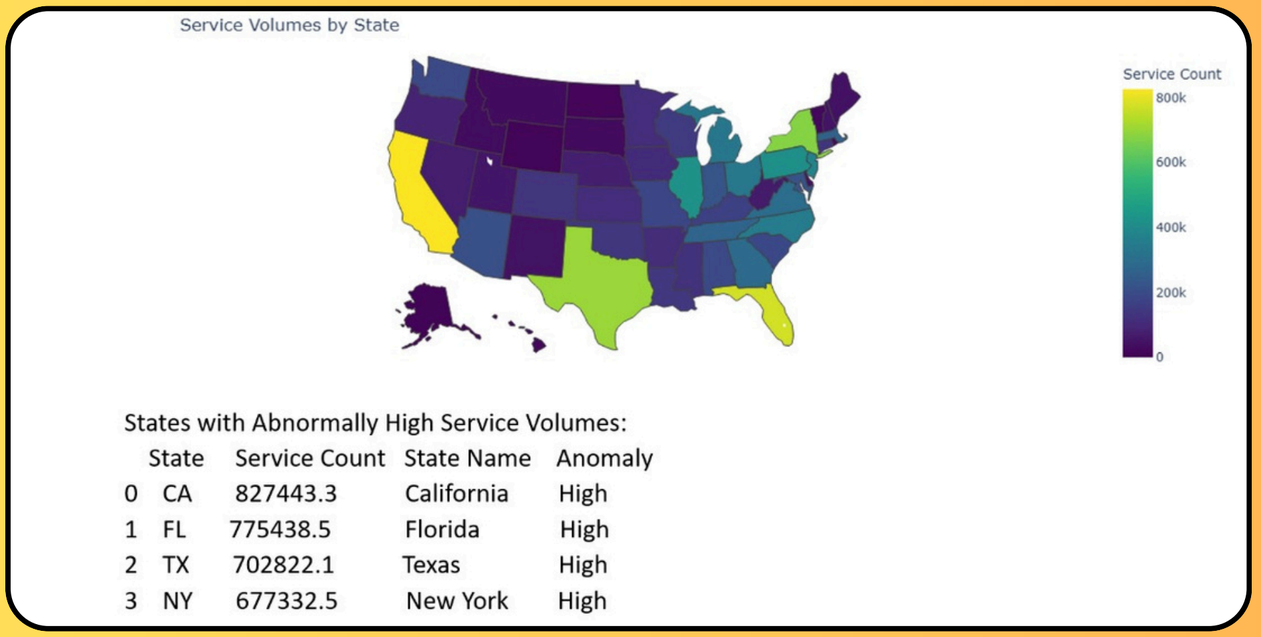
- **Data insights:**

1. **Rows and Columns:** 100000 Rows and 27 Columns.
2. **Key features:** National Provider Identifier, Street Address of the Provider, Zip Code of the Provider and HCPCS Code.
3. **Analysis:** Covers diverse Provider Types and Medicare Payments for robust analysis.

EDA



- **Descending Order:** Providers are ranked by total Medicare earnings, with bars arranged from highest to lowest for easy comparison.
- **Top Earners:** The top two providers (e.g., City of Salem and Area Services, Inc.) significantly outpace others in earnings.
- **Close Competition:** Providers ranked 6th to 10th have smaller differences in earnings, reflecting tighter competition in the lower ranks.

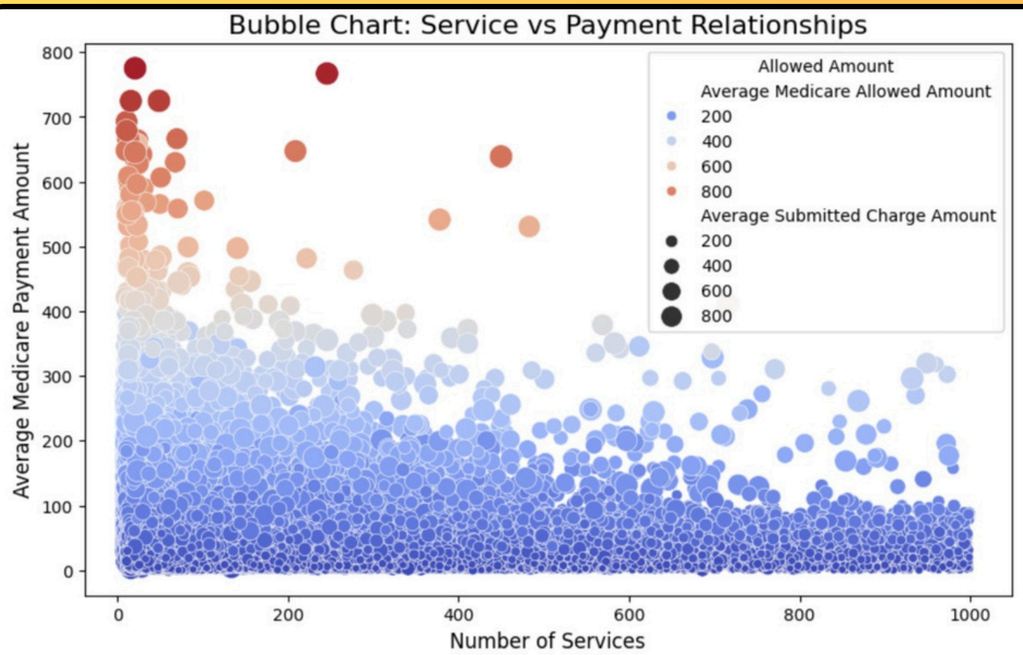


States with Abnormally High Service Volumes:

	State	Service Count	State Name	Anomaly
0	CA	827443.3	California	High
1	FL	775438.5	Florida	High
2	TX	702822.1	Texas	High
3	NY	677332.5	New York	High

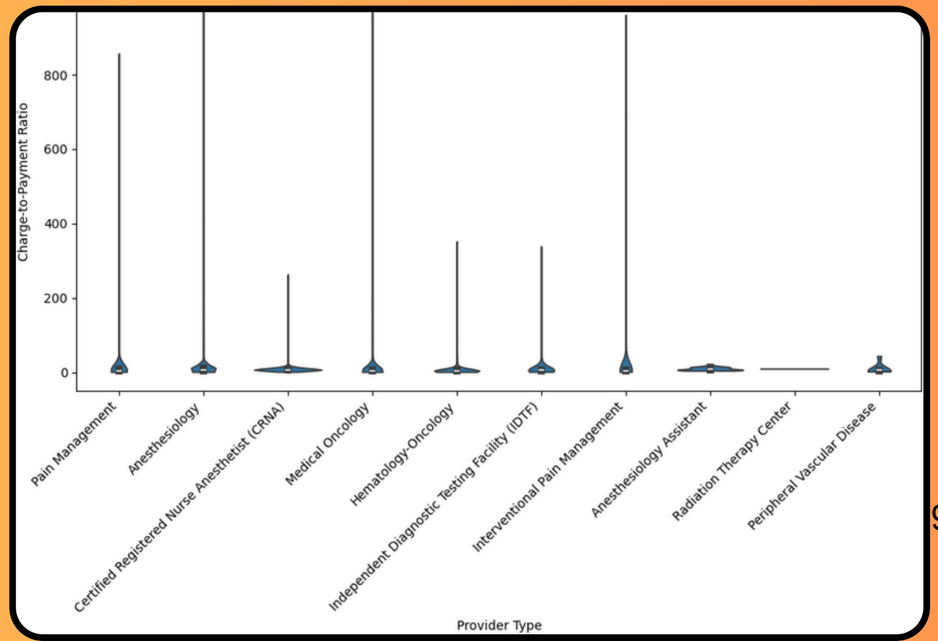
- **Service Volume Distribution**: The map visually represents service counts across U.S. states, with darker shades indicating lower volumes and lighter shades highlighting higher service volumes.
- **Top States**: California, Florida, Texas, and New York lead with significantly high service volumes, exceeding 600,000 each.
- **Regional Anomalies:** These four states show a stark contrast to other regions, underscoring their disproportionate contribution to service volumes nationwide.

EDA



- **Service-Payment Trend**: The chart shows a general pattern where higher Medicare payment amounts are associated with fewer services, while lower payments correspond to a higher number of services.
- **Bubble Size and Color**: Larger and darker bubbles indicate higher "Allowed" or "Submitted" amounts, emphasizing outliers with significant payments or charges.
- **Diverse Distribution**: Most data points cluster at the lower end of the payment scale, but notable outliers exist in the top-left region, representing cases with high average payments and fewer services

- **High Variability in Ratios:** Provider types like Anesthesiology have extreme outliers with charge-to-payment ratios exceeding 1000, indicating significant variability.
- **Tighter Distributions:** Categories like Peripheral Vascular Disease and Radiation Therapy Center show more consistent ratios, suggesting predictable billing and reimbursement practices.
- **Skewed Distributions:** Most provider types have a low interquartile range (IQR) but are affected by outliers, highlighting disparities in billing practices and potential areas for investigation.



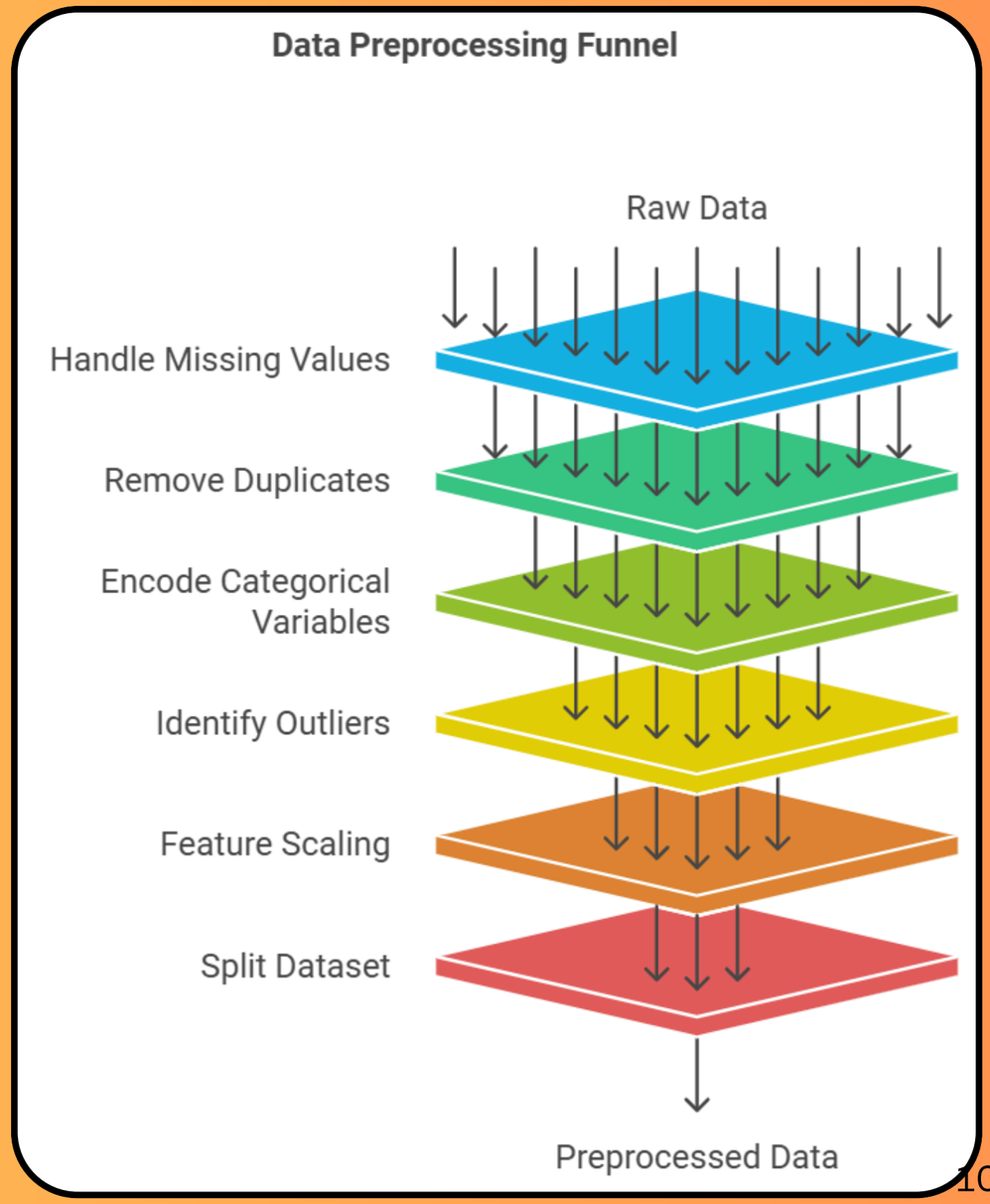
Data Preprocessing

- **Feature Engineering:**

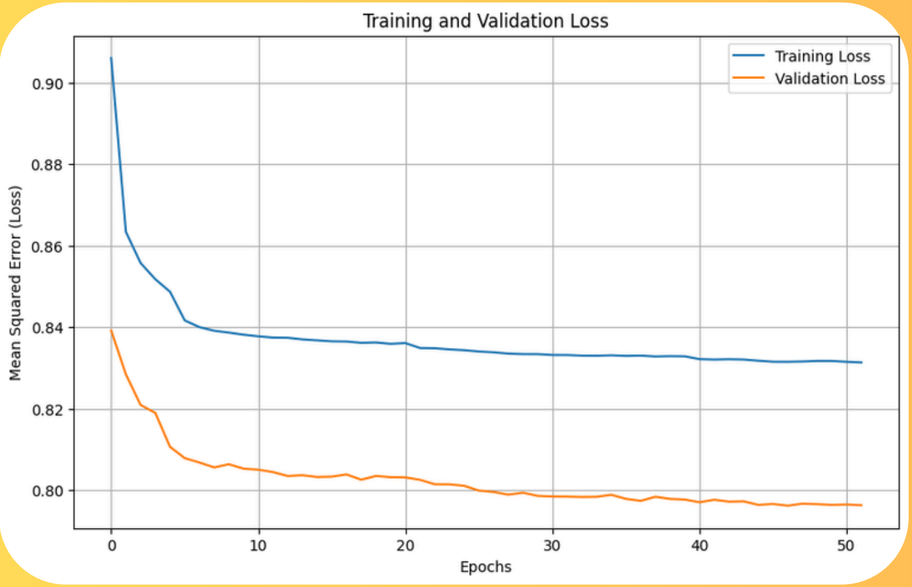
- **Outlier Handling:** Applied capping to maintain data integrity.

- **Data Transformation:**

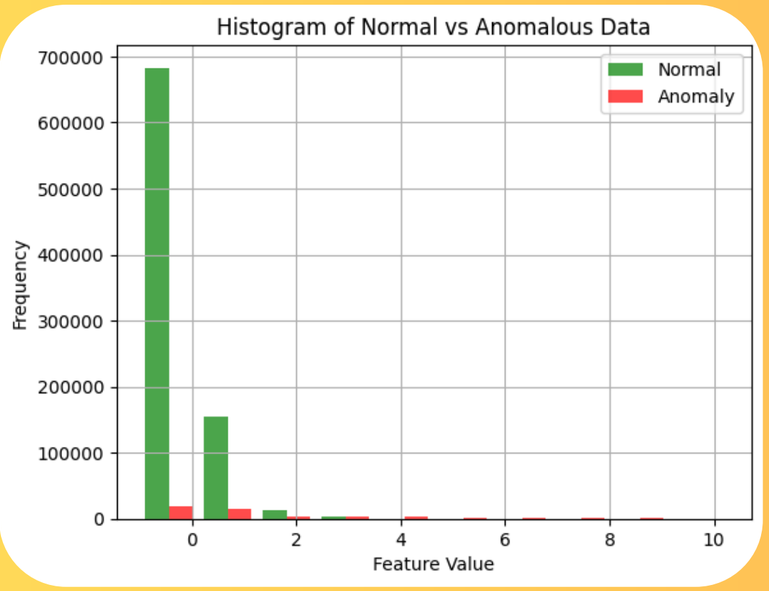
1. MinMax Scaling for numerical features.
2. Ordinal Encoding for ranked data.
3. OneHot Encoding for categorical variables.
4. Label Encoding for identifiers.



Model Building & Evaluation



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User Interface

Challenges and Scopes

Challenges in the Dataset

- Missing Data: Key fields like "Provider First Name" (4,255 missing) and "Street Address 2" (59,363 missing) have significant gaps.
- Data Consistency: "Provider Zip Code" in floating-point format may hinder geospatial analysis.
- Data Imbalance: Dominance of certain categories like "HCPCS Code" can skew results.

Current Scope

- Provider Analysis: Evaluate top performers using metrics like "Number of Services" and "Average Medicare Payment."
- Cost Efficiency: Analyze gaps between "Submitted Charge" and "Medicare Allowed" for optimization.
- Regional Insights: Assess state/city data for accessibility and underserved areas.

Future Scope

- Fraud Detection: Real-time models to flag fraudulent claims.
- Predictive Analytics: Forecast demand and provider performance using historical data.
- Data Enrichment: Add demographic and socio-economic context for deeper insights.





Thank You!

From team

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